

BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Institutional Executive Report

PRIVATE APPLICATION DRAFT - v0.2.3

A clean institutional report preserving the full BRAVO evidence stack: Brazilian equity risk, volatility transmission, synthetic option-overlay logic, BSTI policy selection, and portfolio-governance diagnostics.

CREATED BY

Rodolfo Pereira

Research, portfolio risk, derivatives overlays, and market transmission analysis

Signal date: 2026-06-09 - Not investment advice

EXECUTIVE MEMO

Front-office decision read

BRAVO Lab Front-Office Executive Memo

Purpose: portfolio-governance decision support for Brazilian risk, volatility transmission, and overlay policy selection.

Decision read

BRAVO Lab currently reads Brazilian risk through a BSTI score of 15.17, classified as Fragile, with Brazil Drawdown Pressure as the dominant pressure channel. The current BSTI policy action is Covered Call.

The model is prioritizing income capture. The committee should check whether the upside sold is acceptable under the current stress state.

Portfolio action snapshot

Item	Current read
Current BSTI score	15.17
Current BSTI regime	Fragile
Dominant pressure channel	Brazil Drawdown Pressure
Current policy action	Covered Call
Dominant historical policy choice	Passive Brazil Equity (60.93%)
BSTI policy annualized active return	5.54%
BSTI policy tracking error	10.18%
BSTI policy information ratio	0.54
BSTI policy max drawdown	-25.01%

Evidence stack

Question	Evidence read
Which strategy had the best information ratio?	Bsti Policy Overlay
Which strategy had the best drawdown profile?	Collar
Which strategy had the highest annualized return?	Bsti Policy Overlay
How persistent are warning states?	Average warning duration: 1.89 observations
How persistent are stress states?	Average stress duration: 1.71 observations
How often do warnings escalate?	Warning-to-stress escalation rate: 24.14%
Which BSTI calibration is strongest?	Balanced, 63d horizon, threshold 10.00, governance score 0.50

Risk committee agenda

1. Confirm whether the current BSTI state is persistent or only a temporary pressure print.
2. Check whether the implied policy action matches the committee's drawdown tolerance.
3. Compare the BSTI policy overlay against passive equity, covered calls, collars, and the local stress-aware overlay.

4. Review whether the selected action is justified after transaction costs, liquidity, taxes, and implementation constraints.

5. Decide whether the stress state requires monitoring, hedge discussion, or actual overlay activation.

Implementation warning

This memo is a decision-support layer, not an investment recommendation. The evidence is generated from public market data, synthetic option-premium logic, transparent stress classification, and reproducible CSV outputs. Real B3 option-chain data, liquidity filters, tax effects, and portfolio mandate constraints must be added before production use.

Visual Evidence

See reports/figures/ for the executive dashboard, drawdown map, BSTI signal, policy-selection mix, risk-return map, transition matrix, and calibration scorecard.

EXECUTIVE SYNTHESIS

What this report must prove

BRAVO Lab is not a visual dashboard and it is not a trading promise. It is a reproducible research workflow that converts Brazilian market stress into a portfolio-governance discussion. The project's value comes from the connection between market state, stress classification, synthetic overlay behavior, active-risk diagnostics, BSTI threshold validation, and explicit implementation limits.

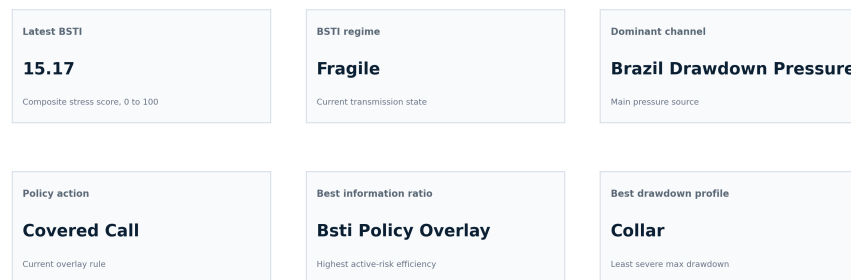
The correct reading is therefore evidence-first. If the current signal is fragile and the policy action is covered call, the immediate question is not whether the model is right in an abstract sense. The question is whether the portfolio committee accepts income capture at the cost of selling part of the upside under the current stress state. If the best drawdown profile is the collar, the next question is whether downside control is worth the protection cost. If the BSTI policy overlay has the strongest information ratio, the reviewer must ask whether that active deviation survives transaction costs, liquidity, taxes, and real B3 option-chain validation.

This version keeps the visual discipline of the clean PDF, but restores the real substance from the original report: data provenance, regime diagnosis, baseline risk, synthetic overlay results, active risk by regime, multi-asset stress signals, BSTI validation, calibration, policy selection, persistence, option attribution, and implementation warnings.

Executive risk dashboard

BRAVO Lab Executive Risk Dashboard

Brazilian risk, allocation, volatility, options, and stress-transmission intelligence



Source: BRAVO Lab processed outputs. Dashboard is generated automatically from reproducible CSV evidence files.

Figure: 00_executive_risk_dashboard.png

The dashboard is the fastest view of the current BRAVO state. It should be read together with the front-office memo, not in isolation. Its job is to summarize the current BSTI signal, regime, dominant pressure channel, policy action, and leading risk diagnostics before the reader moves into the evidence stack.

PORTFOLIO QUESTION

3. Portfolio Question

This section restores the core question from the original report. The project is not asking which strategy is universally best. It is asking which action is defensible under a specific Brazilian market regime.

The core question is not whether passive equity, covered calls, collars, or stress-aware switching are better in isolation. The correct question is regime-dependent.

When Brazilian equity risk changes state, should the portfolio keep full beta, sell volatility through covered calls, pay for downside protection through collars, or switch exposure based on regime evidence?

This report gives the first reproducible answer. It measures the market state, classifies the volatility and drawdown regime, compares synthetic overlays, calculates active risk versus passive Brazilian equity, and turns the result into a portfolio decision frame.

MARKET STATE

4. Market State

The market-state layer gives the cross-market context behind the overlay discussion. It is the bridge between local Brazilian equity behavior and broader risk pressure.

The baseline market layer tracks Brazilian equity exposure, external Brazil exposure, global equity risk, USD/BRL, and VIX. This gives a first cross-market view of local beta, external Brazil risk, global risk appetite, currency stress, and volatility pressure.

This is not yet a full production model. It is a clean research base. The value is transparency. A reviewer can inspect the assumptions, rerun the output, and challenge the decision logic.

DATA PROVENANCE AND EVIDENCE CLASSIFICATION

5. Data Provenance and Evidence Classification

This is one of the most important institutional sections. It separates observed public market data, model-derived metrics, synthetic derivative assumptions, and rule-based strategy logic.

A robust report must separate observed data from modeled signals.

The market layer uses real public market data downloaded through Yahoo Finance with yfinance. The derivatives layer is synthetic. Covered call and collar premiums are estimated through the Black-Scholes engine until real B3 listed-option chains are integrated.

This distinction matters. A real price series can support risk measurement. A synthetic option premium can support research design. It cannot yet support live execution decisions.

Layer	Item	Source	Evidence Type	Status
market_data	brazil_equity	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	brazil_external	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	global_equity	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	fx_usdbrl	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	vix	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
derived_metric	realized_volatility	BRAVO Lab calculation	model_derived	derived from real market data

Layer	Item	Source	Evidence Type	Status
derived_metric	drawdown	BRAVO Lab calculation	model_derived	derived from real market data
regime_signal	baseline_regime_classifier	BRAVO Lab rule-based classifier	model_generated_signal	decision signal, not observed market data
synthetic_derivatives	covered_call_overlay	BRAVO Lab synthetic option engine	synthetic_research_assumption	not real B3 option-chain evidence
synthetic_derivatives	collar_overlay	BRAVO Lab synthetic option engine	synthetic_research_assumption	not real B3 option-chain evidence
strategy_logic	stress_aware_overlay	BRAVO Lab switching rule	model_generated_strategy_rule	research rule requiring validation

REGIME DIAGNOSIS

6. Regime Diagnosis

The regime diagnosis gives the baseline state of the market before more complex stress logic is added. This is the first checkpoint for deciding whether overlay behavior should be passive, income-oriented, or protection-oriented.

The current Brazilian equity signal sits in stress.

That matters because the same overlay can behave well in one regime and poorly in another. A covered call can create useful income in a range-bound market, but it can also sell the recovery too cheaply. A collar can protect the book during stress, but it can also become expensive insurance if drawdown risk fades.

The regime classifier uses realized volatility, volatility percentile, and drawdown. It is simple by design. The point is to create an auditable baseline before adding GARCH, MTV-GARCH, stress transmission indexes, wavelets, CCA, or machine learning.

BASELINE RISK METRICS

7. Baseline Risk Metrics

The baseline risk table is not a final allocation model. It is the risk map that tells the reader what kind of environment the overlay logic is operating inside.

The table below gives the first risk layer across the monitored assets. It is not a final allocation model. It is the risk map used to decide whether the overlay discussion is taking place in a calm, fragile, or stressed environment.

Asset	Ann. Return	Ann. Volatility	Sharpe	Sortino	Max Drawdown	VaR 95%	CVaR 95%	Obs.
brazil_equity	9.92%	23.14%	0.525	0.686	-46.93%	-2.21%	-3.28%	3239
fx_usdbrl	6.34%	16.45%	0.456	0.703	-26.80%	-1.59%	-2.23%	3239
brazil_external	2.57%	33.91%	0.246	0.323	-66.54%	-3.29%	-4.84%	3239
global_equity	13.30%	16.92%	0.823	0.985	-33.72%	-1.60%	-2.59%	3239
vix	2.24%	133.44%	0.624	1.276	-85.66%	-10.52%	-14.20%	3239

SYNTHETIC OVERLAY RESULTS

8. Synthetic Overlay Results

This section must remain explicit about the synthetic nature of the option engine. Covered calls and collars are research overlays until observed B3 option-chain data is integrated.

The first overlay engine compares four exposures:

- passive Brazilian equity exposure
- synthetic covered call overlay
- synthetic protective collar overlay
- stress-aware overlay switching

The engine uses a 21-trading-day rebalance approximation, synthetic Black-Scholes option premiums, and a transaction-cost assumption of 5.0 basis points per option leg.

The stress-aware overlay maps regimes into actions: passive exposure in calm conditions, covered call income in fragile conditions, and collar protection in stress or extreme-stress conditions. It does not claim live tradability. It is a controlled research baseline before adding real B3 option chains, taxes, liquidity, and execution constraints.

Returns in this section are approximately monthly strategy-period returns, annualized using 12.0 periods per year.

Strategy	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Best Period	Worst Period	Obs.
passive_brazil_equity	9.93%	22.24%	0.543	-37.77%	21.22%	-35.92%	151
covered_call	7.92%	16.80%	0.549	-36.79%	10.57%	-35.16%	151
collar	5.54%	10.23%	0.579	-21.49%	4.11%	-4.76%	151
stress_aware_overlay	6.79%	13.62%	0.552	-23.01%	10.58%	-14.33%	151

ACTIVE RISK DIAGNOSTICS

9. Active Risk Diagnostics

Absolute return is not enough. This section restores the active-risk layer: tracking error, information ratio, hit rate, downside hit rate, and worst active period versus passive Brazilian equity.

Absolute return is not enough. A portfolio desk also needs to know whether the overlay earns enough to justify its deviation from passive Brazilian equity.

Tracking error measures how far the overlay moves away from passive exposure. The information ratio tests whether that active deviation is rewarded. Hit rate shows how often the overlay beats passive. Downside hit rate is stricter. It asks whether the overlay helps when passive exposure is already losing money.

Strategy	Ann. Active Return	Tracking Error	Information Ratio	Hit Rate	Downside Hit Rate	Worst Active Period	Obs.
covered_call	-2.87%	10.33%	-0.278	74.83%	100.00%	-16.46%	151
collar	-6.16%	14.82%	-0.416	69.54%	100.00%	-17.62%	151
stress_aware_overlay	-4.57%	13.62%	-0.335	44.37%	64.18%	-17.62%	151

ACTIVE RISK BY REGIME

9. Active Risk by Regime

This is one of the most important portfolio-governance sections. It asks whether an overlay that looks acceptable across the full sample remains governable in calm, fragile, stress, and extreme-stress states.

Full-sample tracking error is useful, but it is not enough for portfolio governance. A strategy can look acceptable across the whole sample while creating too much active risk in a specific market state.

This diagnostic shows where each overlay creates tracking error versus passive Brazilian equity exposure: calm markets, fragile markets, stress markets, or extreme-stress markets.

Regime	Strategy	Annualized Active Return	Tracking Error	Information Ratio	Hit Rate	Downside Hit Rate	Avg. Active Period	Best Active Period	Worst Active Period	Obs.
calm	covered_call	-17.41%	10.71%	-1.625	56.67%	100.00%	-1.45%	1.70%	-7.93%	30
calm	collar	-23.87%	10.63%	-2.246	50.00%	100.00%	-1.99%	0.59%	-8.93%	30
calm	stress_aware_overlay	-10.70%	9.17%	-1.167	26.67%	57.14%	-0.89%	1.70%	-8.23%	30
extreme_stress	covered_call	1.49%	13.47%	0.111	85.29%	100.00%	0.12%	7.34%	-16.46%	34
extreme_stress	collar	7.39%	25.25%	0.293	82.35%	100.00%	0.62%	31.28%	-17.62%	34
extreme_stress	stress_aware_overlay	3.31%	24.49%	0.135	64.71%	75.00%	0.28%	31.28%	-17.62%	34
fragile	covered_call	-1.33%	7.28%	-0.183	76.19%	100.00%	-0.11%	1.84%	-6.06%	21
fragile	collar	-6.61%	7.71%	-0.858	76.19%	100.00%	-0.55%	2.20%	-6.99%	21
fragile	stress_aware_overlay	-2.02%	5.09%	-0.397	38.10%	55.56%	-0.17%	2.20%	-3.53%	21
neutral	covered_call	-4.74%	8.09%	-0.587	66.67%	100.00%	-0.40%	2.84%	-7.95%	21
neutral	collar	-7.34%	10.12%	-0.725	66.67%	100.00%	-0.61%	4.10%	-8.91%	21
neutral	stress_aware_overlay	-8.34%	7.37%	-1.131	9.52%	11.11%	-0.70%	1.03%	-8.91%	21
stress	covered_call	3.69%	9.01%	0.410	82.22%	100.00%	0.31%	2.52%	-13.39%	45
stress	collar	-3.84%	9.60%	-0.400	71.11%	100.00%	-0.32%	4.20%	-14.07%	45
stress	stress_aware_overlay	-5.86%	9.14%	-0.641	60.00%	81.82%	-0.49%	1.83%	-14.07%	45

Cumulative performance path

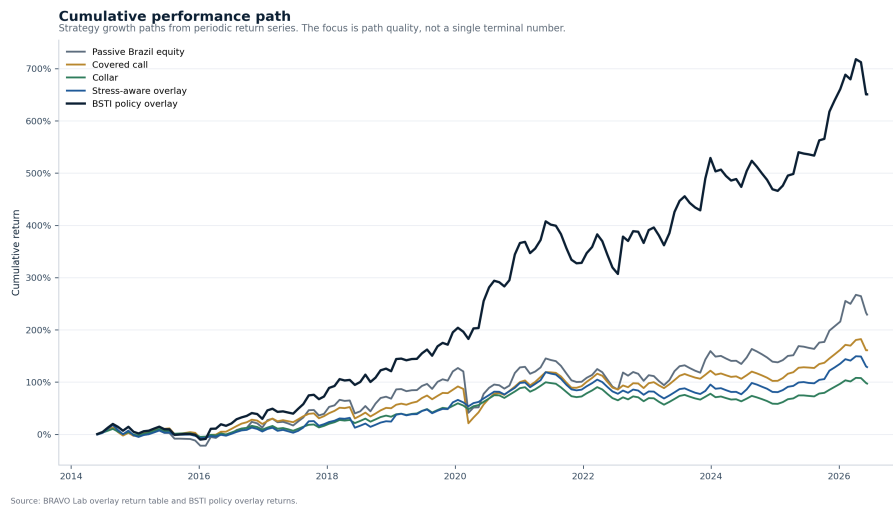


Figure: 01_cumulative_performance.png

The cumulative performance chart should not be read as a simple winner-takes-all ranking. It shows how the tested exposures travel through time. The professional question is what kind of risk, drawdown, tracking error, and implementation burden produced each line.

Drawdown control map

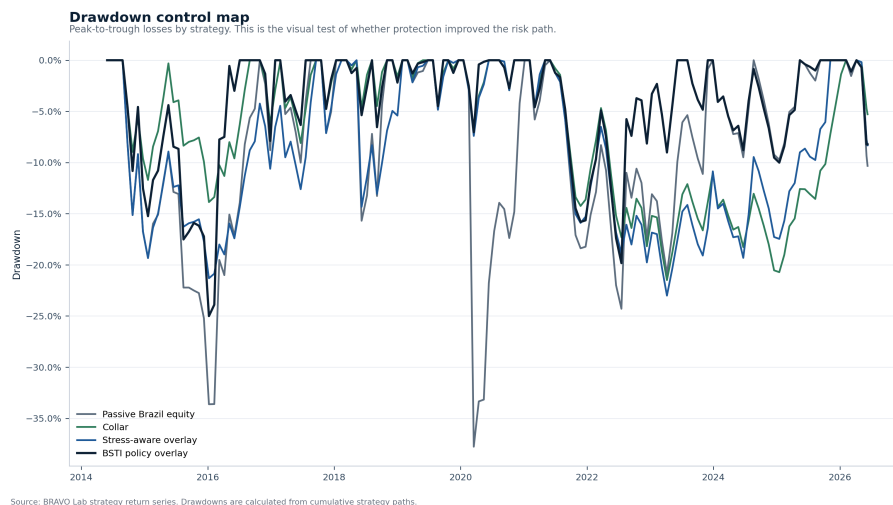


Figure: 02_drawdown_profile.png

The drawdown chart is where the overlay logic earns or loses credibility. A strategy can look acceptable in performance space and still fail when the portfolio is under pressure. The collar may look less exciting in return terms but can matter if drawdown control is the committee's priority.

MULTI-ASSET STRESS SIGNALS

10. Multi-Asset Stress Signals

This restores the broader stress layer beyond local Brazilian equity behavior. BRAVO is strongest when it identifies whether pressure is coming from local drawdown, FX, external Brazil exposure, global equity risk, or VIX.

The first regime layer is built from local Brazilian equity behavior. That is useful, but incomplete. A portfolio stress system should also look outside the local price series.

This section adds external Brazil exposure, global equity pressure, USD/BRL pressure, and VIX pressure to create a broader multi-asset stress dashboard. It does not forecast markets. It identifies where pressure is currently coming from.

Date	Composite Stress Score	Stress Regime	Top Pressure 1	Value 1	Top Pressure 2	Value 2	Top Pressure 3	Value 3	Brazil Drawdown	Brazil 21D Vol	VIX Level
2026-06-09	0.401	calm	brazil_drawdown_pressure	1.877	fx_pressure	0.303	global_equity_pressure	0.191	-15.20%	16.15%	18.920

BRAZIL STRESS TRANSMISSION INDEX

Brazil Stress Transmission Index

The BSTI converts individual stress inputs into a formal composite index from 0 to 100. The purpose is not prediction. The purpose is monitorable portfolio governance.

![Brazil Stress Transmission Index](figures/03_bsti_signal.png)

Interpretation: Displays warning and stress zones so the stress signal becomes interpretable as a governance state.

Brazil Stress Transmission Index chart

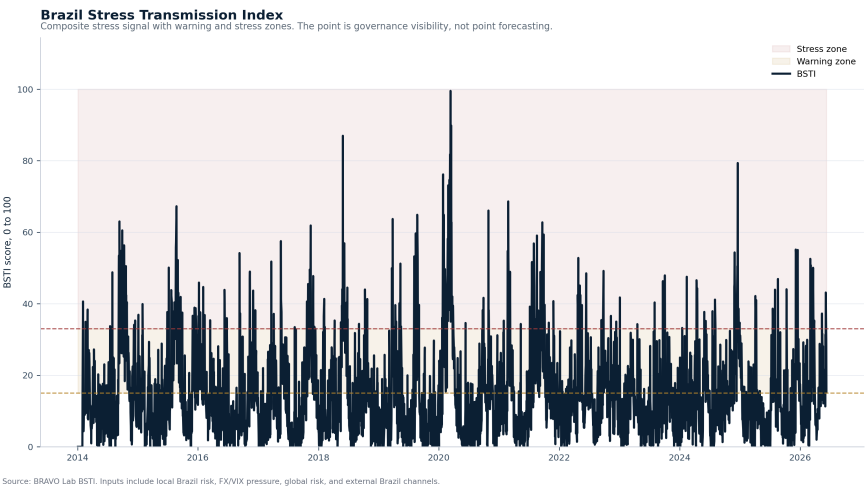


Figure: 03_bsti_signal.png

The BSTI chart makes the stress state visible across time. It should be read as a governance signal with warning and stress zones, not as a standalone forecast.

BSTI THRESHOLD VALIDATION

12. BSTI Threshold Validation

This section is essential. A stress index is only useful if its thresholds connect to portfolio-relevant outcomes such as future drawdowns and overlay behavior under elevated stress.

A stress index is only useful if its thresholds connect to portfolio-relevant outcomes. This validation layer tests whether BSTI levels are associated with future benchmark losses, future drawdowns, and overlay behavior when stress is already elevated.

This does not claim forecasting power. It tests whether BSTI is informative enough to support portfolio-governance discussion.

BSTI THRESHOLD CALIBRATION

13. BSTI Threshold Calibration

Calibration is where the prototype becomes auditable. The purpose is not overfitting. The purpose is to test whether different thresholds and channel weights create more useful governance signals.

The first BSTI threshold validation tests a fixed index structure. This calibration layer asks a more demanding question: does the stress index behave better under alternative thresholds and channel weights?

The point is not to overfit the index. The point is to test whether the chosen structure is robust enough for portfolio-governance discussion.

BSTI calibration scorecard

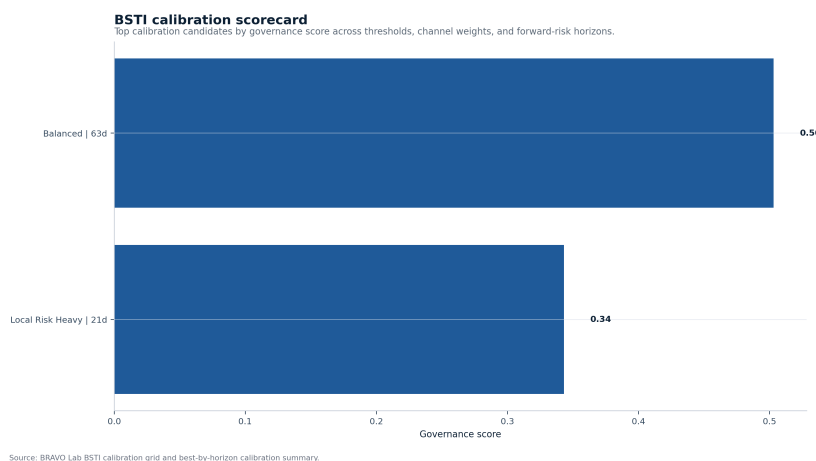


Figure: 07_bsti_calibration_scores.png

The calibration scorecard should be read as part of the validation layer. A strong candidate is not proof of forecasting power; it is evidence that the stress index can be tested, compared, and improved.

BSTI OVERLAY POLICY SELECTION

14. BSTI Overlay Policy Selection

This is where BRAVO moves from diagnosis to portfolio action. The policy layer converts low, medium, and high stress into passive exposure, covered-call income, or collar protection.

The BSTI policy layer converts the stress index into an explicit portfolio action. This is where the project moves from diagnosis to governance.

The rule is intentionally simple:

- low BSTI: remain in passive Brazil equity exposure
- medium BSTI: use covered calls to collect option premium
- high BSTI: use collars to prioritize downside control

This does not claim to forecast returns. It tests whether a transparent stress signal can discipline overlay selection across passive exposure, covered calls, collars, and the existing local-regime stress-aware overlay.

BSTI policy-selection mix

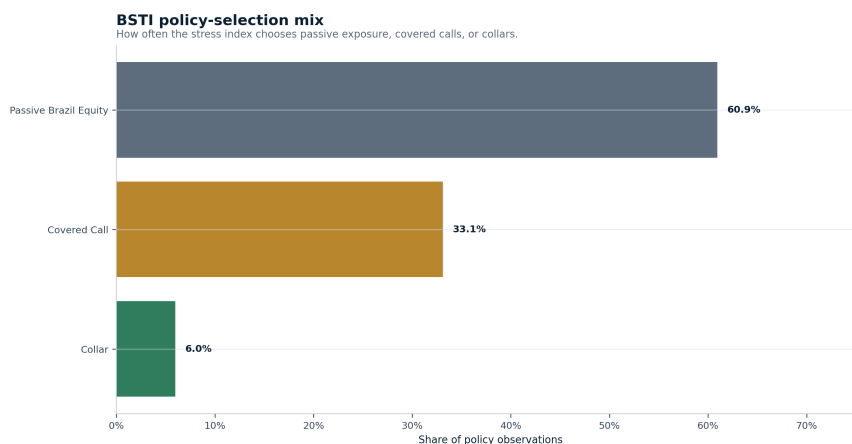


Figure: 04_bsti_policy_selection_mix.png

The policy mix shows whether the system is mostly passive, income-oriented, or defensive. A useful policy rule must change behavior only when the evidence deserves it.

BSTI SIGNAL PERSISTENCE

15. BSTI Signal Persistence and State Transitions

Persistence matters because a temporary warning does not deserve the same response as a signal that tends to remain fragile or escalate into stress.

A stress signal is not useful only because it moves. It is useful when it persists long enough to support a decision process.

This section tests whether the Brazil Stress Transmission Index behaves like a governance signal rather than a random dashboard number. It classifies each BSTI observation into normal, warning, or stress states, then studies transition probabilities, state duration, warning-to-stress escalation, and dominant pressure-channel rotation.

BSTI state-transition matrix

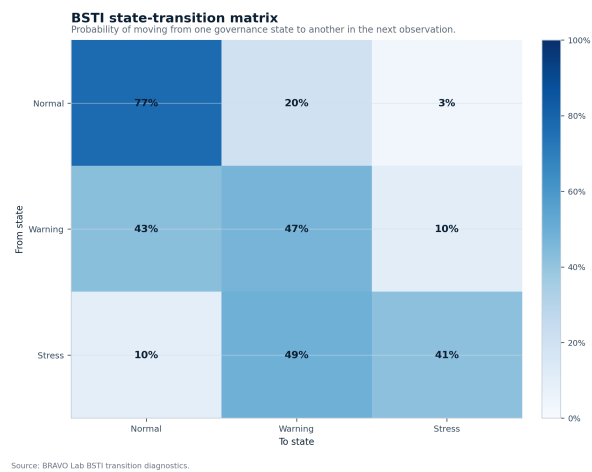


Figure: 06_bsti_transition_matrix.png

The transition matrix helps determine whether stress states behave like persistent warning processes or random flashes. This is central to deciding whether action is justified.

DRAWDOWN AND RECOVERY DIAGNOSTICS

16. Drawdown and Recovery Diagnostics

This section tests what happens after loss. It matters because overlays are judged not only by how they behave during drawdown, but also by whether they damage the recovery path.

Drawdown diagnostics ask whether the overlay helps at different levels of benchmark pain. Recovery diagnostics ask the uncomfortable second question: does the hedge damage the portfolio when the market starts rebounding?

This matters because a protective overlay can be useful during the fall and still become expensive during the recovery.

STRATEGY HELP-HURT DIAGNOSTICS

18. Strategy Help-Hurt Diagnostics

This section matters because a strategy can help in one context and hurt in another. The report should preserve that distinction instead of summarizing everything into one average.

The help-hurt diagnostic explains the trade-off behind each overlay. A strategy can protect the downside and still hurt the portfolio if it gives away too much rebound. It can also improve income while failing to protect the book when the benchmark is already falling.

This section separates the periods where each overlay adds active value from the periods where it creates drag versus passive Brazilian equity.

Strategy	Avg. Active Return	Active When Passive Up	Active When Passive Down	Hit Rate	Missed Upside Rate	Downside Protection Rate	Best Active Period	Worst Active Period	Primary Help	Primary Hurt	Obs.
covered_call	-0.24%	-1.32%	1.08%	74.83%	45.78%	100.00%	7.34%	-16.46%	downside_protection	missed_upside	151
collar	-0.51%	-2.23%	1.60%	69.54%	55.42%	100.00%	31.28%	-17.62%	downside_protection	missed_upside	151
stress_aware_overlay	-0.38%	-1.61%	1.13%	44.37%	38.55%	64.18%	31.28%	-17.62%	downside_protection	missed_upside	151

IMPLEMENTATION DRAG DIAGNOSTICS

19. Implementation Drag Diagnostics

This is where the project becomes more honest. Gross signal quality is not enough. Real implementation requires costs, frictions, liquidity, taxes, and execution assumptions.

The implementation-drag diagnostic separates the gross overlay signal from the net result after transaction costs. This matters because an overlay can look useful before costs and still fail after realistic rebalancing drag.

The diagnostic does not yet decompose every option leg into premium income, payoff effect, and moneyness attribution. It is the intermediate institutional layer showing whether the overlay signal survives implementation.

Strategy	Gross Active	Implementation Drag	Net Active	Total Drag	Drag / Gross Signal	Cost Survival	Active When Passive Up	Active When Passive Down	Best Net Active	Worst Net Active	Obs.
covered_call	-0.19%	-0.05%	-0.24%	-7.55%	0.265	1.265	-1.32%	1.08%	7.34%	-16.46%	151
collar	-0.41%	-0.10%	-0.51%	-15.10%	0.242	1.242	-2.23%	1.60%	31.28%	-17.62%	151
stress_aware_overlay	-0.32%	-0.06%	-0.38%	-8.85%	0.182	1.182	-1.61%	1.13%	31.28%	-17.62%	151

OPTION OVERLAY ATTRIBUTION

20. Option Overlay Attribution

Option attribution separates premium, protection cost, payoff, drag, and net effect. This is critical because covered calls and collars are not single objects. They are bundles of economic components.

This attribution layer explains why the synthetic option overlay worked or failed. It separates the model-implied return into premium income, protection cost, payoff effect, implementation drag, and net overlay effect.

This is not real B3 option-chain attribution. It is model-implied attribution based on the synthetic Black-Scholes overlay engine used in this project. That disclosure is important because it keeps the research honest while still making the derivatives logic inspectable.

Strategy	Avg. Underlying Return	Avg. Call Premium Income	Avg. Put Protection Cost	Avg. Option Payoff Effect	Avg. Implementation Drag	Avg. Gross Overlay Effect	Avg. Net Overlay Effect	Avg. Net Strategy Return	Positive Net Overlay Rate	Obs.
collar	1.05%	1.30%	-1.23%	-0.73%	-0.10%	-0.67%	-0.77%	0.28%	24.03%	154
covered_call	1.05%	1.30%	0.00%	-1.63%	-0.05%	-0.33%	-0.38%	0.67%	72.08%	154
stress_aware_overlay	1.05%	0.94%	-0.78%	-0.72%	-0.06%	-0.57%	-0.63%	0.42%	20.78%	154

OVERLAY DECISION MATRIX

22. Overlay Decision Matrix

The decision matrix is where the report should become practical: when each strategy is useful, dangerous, or still unvalidated.

Strategy	Best Use	Main Risk	Portfolio Reading
Passive Brazilian Equity	Clean trend, calm regime, strong rebound	Full downside exposure	Use when upside participation matters more than protection
Covered Call	Fragile regime, elevated volatility, income objective	Upside is capped	Use when premium harvesting matters more than full upside capture

Strategy	Best Use	Main Risk	Portfolio Reading
Collar	Stress regime, drawdown pressure, capital preservation	Protection cost and capped upside	Use when left-tail control matters more than return maximization
Stress-Aware Overlay	Regime-dependent switching	Model risk and signal timing	Uses passive in calm regimes, covered calls in fragile regimes, and collars in stress regimes

NU / NUBANK APPLICATION READ

Why the project is relevant without overselling it

BRAVO Lab should be presented as evidence of applied financial intelligence, not as a finished product. The strongest point is the workflow: data processing, risk diagnostics, derivatives-overlay logic, active-risk measurement, stress transmission, validation, reporting, and clear implementation limits.

For a Nu/Nubank reviewer, this matters because the project shows more than coding. It shows the ability to frame a financial problem, build a reproducible pipeline, compare alternatives, explain risk, and admit what still needs validation. That combination is closer to how real portfolio and risk teams work than a notebook that only shows a high-return line.

The correct application sentence is: BRAVO Lab is a reproducible Python research prototype that converts Brazilian market stress, synthetic derivatives-overlay logic, active-risk diagnostics, and BSTI policy selection into portfolio-governance evidence.

Application layer	Evidence inside BRAVO
Python execution	Scripts, processed data, generated figures, reports, and reproducible outputs.
Brazilian market knowledge	B3 exposure, USD/BRL pressure, external Brazil exposure, VIX, and Brazil-specific stress channels.
Derivatives reasoning	Synthetic covered-call and collar logic, with explicit limits before B3 option-chain integration.
Risk communication	Executive memo, evidence stack, risk committee agenda, and implementation warning.
Research honesty	Clear separation between real market data, model-derived metrics, synthetic assumptions, and strategy rules.

DEEP SWOT

The project is strongest when the limits are part of the argument

The strength of BRAVO Lab is that it is already an end-to-end research system. It has a real decision question, generated evidence, processed outputs, figures, technical sections, and a front-office memo. The weakness is not the idea. The weakness is validation depth, especially around real option chains, liquidity, tax assumptions, and implementation drag.

The opportunity is to evolve BRAVO into a serious evidence portfolio for quant, risk, and portfolio analytics applications. The threat is overclaiming. If the project is framed as a prediction engine, it becomes fragile. If it is framed as a reproducible decision-support prototype, it becomes credible.

SWOT	Institutional read
Strengths	End-to-end Python workflow, Brazil-specific stress lens, derivatives-overlay reasoning, active-risk diagnostics, BSTI framework, and executive communication.
Weaknesses	Synthetic option layer, incomplete implementation friction, limited real B3 option-chain validation, and need for deeper covariance modeling.
Opportunities	Real B3 option chains, MTV-GARCH or DCC, CCA stress compression, wavelet multiscale stress analysis, and stronger Nu/Nubank application positioning.
Threats	Being misunderstood as investment advice, overclaiming predictive power, or reducing the project to visual design without evidence density.

MODEL LIMITS AND EVIDENCE FILES

28. Model Limits and Governance

The final institutional report must preserve the explicit limits from the original version. This is what protects credibility.

This report is intentionally clear about what it does not prove.

- Yahoo Finance data is used as the baseline public-data source.
- No real B3 option-chain data is used yet.
- Option premiums are synthetic Black-Scholes approximations.
- Transaction costs, taxes, spreads, and liquidity constraints are not included yet.
- GARCH and MTV-GARCH are not yet active in this report.
- The regime classifier is transparent but not final.
- Active risk diagnostics are useful but still require stress-window validation.
- This is research infrastructure, not investment advice.

Evidence file	Role
reports/front_office_memo.md	Executive decision read, portfolio action snapshot, evidence stack, risk committee agenda.
reports/baseline_report.md	Full technical and evidence report preserving the original BRAVO logic.
reports/figures/	Generated visual evidence for dashboard, performance, drawdown, BSTI, policy mix, transition matrix, and calibration.
data/processed/	Reproducible CSV outputs behind metrics, tables, and report sections.
scripts/generate_institutional_pdf.py	Current institutional PDF generator.
reports/nu_asset_application_note.md	Recruiter-facing application positioning.

BRAVO Lab

Evidence-first. Clean design. No investment advice.

From Brazilian market stress to portfolio-governance review.

Created by Rodolfo Pereira